INCH-POUND

MIL-W-40145C
11 October 1991
SUPERSEDING
MIL-W-40145B
27 March 1969

MILITARY SPECIFICATION

WRENCHES, CROWFOOT, SPECIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of crowfoot wrench.

1.2 Classification. Wrenches shall be of the following sizes, as specified (see 6.2).

Size 1 - 3/4 inch
Size 2 - 7/8 inch

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5120

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SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

12-1-17 - Wrench, Crowfoot, Special

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 108 - Steel Bars, Carbon, Cold Finished, Standard Quality
B 456 - Electrodeposited Coatings of Copper Plus Nickel Plus Chromium and Nickel Plus Chromium
B 633 - Zinc on Iron & Steel, Electrodeposited Coatings of
E 18 - Metallic Materials Rockwell Hardness & Rockwell Superficial, Hardness of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)



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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Steel bar. The steel bar used for forging wrenches shall be of any commercial steel composition between 1035 through 1045 inclusive, in accordance with ASTM A 108, or any commerical alloy capable of being heat treated to the hardness specified in 3.3.2.

3.3 Construction.

3.3.1 Forming. The wrenches shall be formed to the dimensions shown on Drawing 12-1-17 and in accordance with the manufacturer's current commercial practice.

3.3.2 Hardening. The wrenches shall be heat treated to a hardness of Rockwell C40 to C45 when tested as specified in 4.5.1.

3.4 Torque performance. The wrenches shall be capable of withstanding a torque of 720 inch-pounds for the size 1 wrench and a torque of 1200 inch-pounds for the size 2 wrench without deformation when tested as specified in 4.5.2.

3.5 Finish. The finished wrenches shall be nickel-chromium plated in accordance with service condition 3 of ASTM B 456.

3.6 Marking for identification. The wrenches shall be permanently marked with the manufacturer's name or trademark of such known characters as to be easily identifiable.

3.7 Workmanship. The finished wrenches shall be clean and free from sharp or rough edges, burrs, pits, cracks, rust, and scale. The wrenches shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and shall be tested for the characteristics specified in 4.4.4. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.



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4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of wrenches. The sample unit shall be one wrench. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

TABLE I. End item visual defects

| Examine | Defect |
|---|---|
| Finish (plating) | Not nickel-chromium plating<br>Rust or scale on surfaces<br>Area of no plating or non-adherent |
| Construction and workmanship | Fractured, broken, cracked, or malformed<br>Sharp or rough edges, burrs, or projections |
| Identification marking | Missing, illegible, incorrect, or incomplete |

4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawing. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of wrenches. The sample unit shall be one wrench. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 End item testing. The wrench shall be tested as specified in 4.5.1 and 4.5.2. The lot size shall be expressed in units of wrenches. The sample unit shall be one wrench. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging inspection. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of PPP-P-40.

4.5 Methods of inspection.

4.5.1 Hardness test. Hardness readings shall be made at both ends and in the center of the wrench in accordance with ASTM E 18. The average of the values obtained shall be evaluated to determine compliance with the requirements of 3.3.2. Any noncompliance shall constitute a defect.

4.5.2 Torque test. The wrenches shall be tested on adapters of the applicable size required to fit the wrench. The wrench shall be placed on the adaptor attached to a torque tester. Force shall be applied to the wrench to



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transmit the required test load between the wrench and the restrained adaptor. This shall be done a total of three times on each adaptor, rotating the wrench 30° to the next gripping position of the wrench after each torque cycle to determine compliance with 3.4. If the wrench slips on the adaptor before the test load values can be met, the test shall be considered a failure.

## 5. PACKAGING

5.1 Levels A, B, and Commercial. Preservation, packing, and marking for each level of protection shall be in accordance with the applicable requirements of PPP-P-40 (see 6.2).

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The crowfoot wrenches are intended to be used for the purpose of installing and removing the ignition wiring harness nuts on internal combustion engines.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size wrench required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Hand tool
Nut installation/removal

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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Custodians:

Army - GL
Air Force - 99

Review activities:

Army - AL, ME
Air Force - 84

User activity:

Army - MI

Civil agency coordinating activity:

GSA - FSS

Preparing activity:

Army - GL

Project 5120-C107



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-W-40145C | 91 10 11 |

3. DOCUMENT TITLE
WRENCHES, CROWFOOT, SPECIAL

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

| 6. SUBMITTER | | |
|---|---|---|
| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

| 8. PREPARING ACTIVITY | | |
|---|---|---|
| a. NAME<br>U.S. Army Natick RD&E Center | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>508-651-4531 | (2) AUTOVON<br>256-4531 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Natick RD&E Center<br>ATTN: STRNC-UXT<br>Natick, MA 01760-5017 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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